

Graph-Theoretic and Learning-Based Methods for Enhanced Resource Allocation in Securing Interdependent Systems

Mohammad Riyad Al-Eiadeh

Elmore Family School of Electrical and Computer Engineering
Purdue University

Boiler Maker

Under the Supervision of:

Mustafa Abdallah

School of Applied and Creative Computing
Purdue University



Committee Members

- **Co-Chairs:**

- **Dr. Brian S. King**
Elmore Family School of ECE
- **Dr. Mustafa Abdallah**
Department of CIT



- **Committee Members**

- **Dr. Lingxi Li**
Elmore Family School of ECE
- **Dr. Maher Rizkalla**
Elmore Family School of ECE



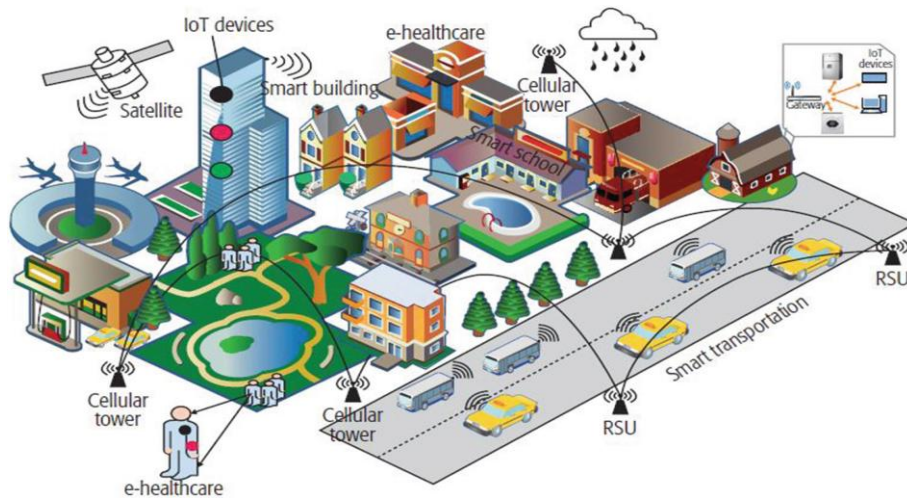
Agenda

- Problem Statement and Thesis Main Contributions
- Overview of Interdependent Systems Framework
- Recall: Preliminary work explored top-attack-path analysis using GA, centrality-driven allocation, meta-learning, and graph embedding / ranking approaches for enhancing CPS security
 - [INTCES24, SATC25, C&S24, IJIS25, TOPS25, TDSC25]
- Meta-Learning based on hand-crafted attack graph features for Automated Resource Allocation [JISA*]
- Stress-testing attack-graph based resource allocation defense schemes under adversarial attacks [TOPS*]
- Future Research Directions

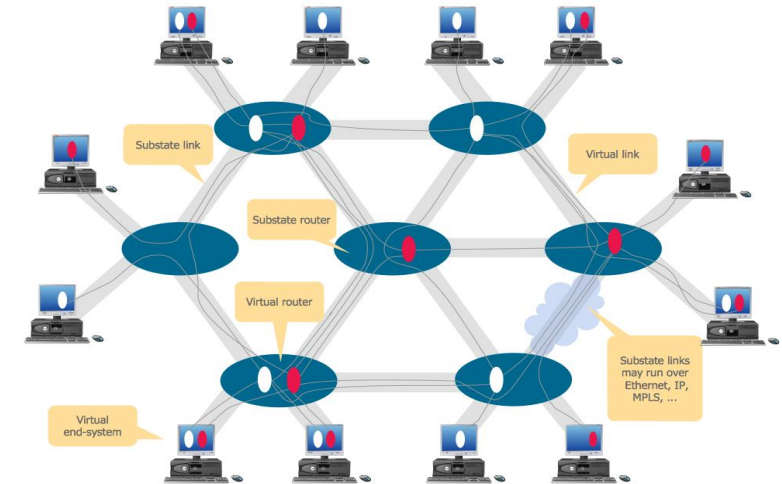
Problem Statement and Thesis Main Contributions

Networked Systems

- Modern systems are networked systems (e.g., Cyber-physical Systems)
 - Integration of **sensing, computation, control and networking into physical objects** and infrastructure
 - Connecting all **objects to the Internet and to each other**



Source: https://link.springer.com/chapter/10.1007/978-3-030-41560-0_14



Source: <https://janeyroselee.blogspot.com/2022/09/10-vlan-network-diagram.html>

Main Challenges in Interdependent Systems

Security Issues

- Rapid Increase in cyber-attacks
- Suboptimal security policy



Source: Center for Strategic & International Studies

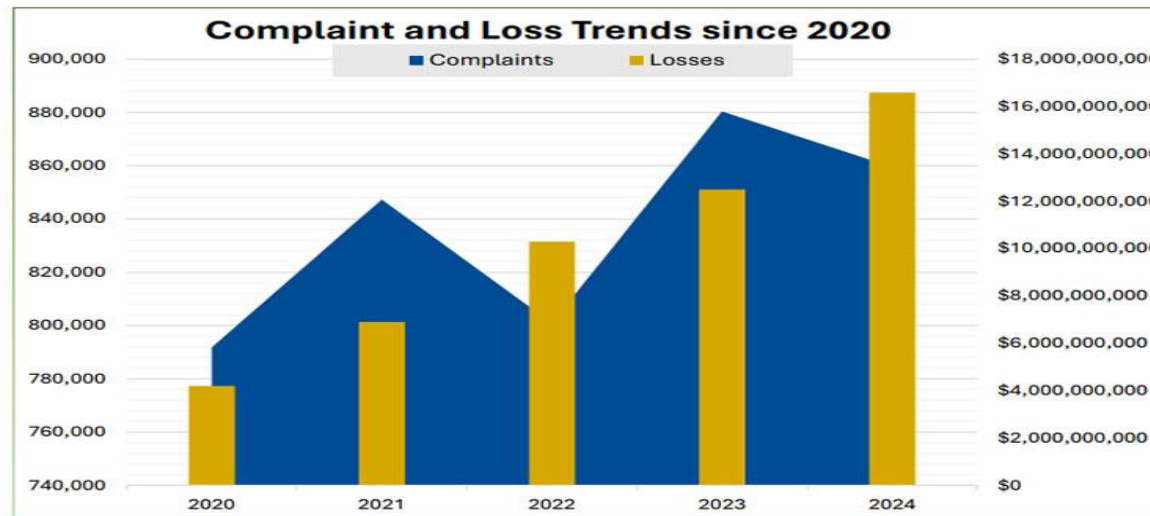
High Costs

- Cost of successful attack
- Cost of security policy



Huge Cost of Real-World Cyber-Attacks

- Statistics reveal the **financial losses** incurred by **US companies** due to **cybercrime** from 2020 to 2024
- Reported by the **Federal Bureau** of Investigation's Internet Crime Complaint Center (IC3)



Literature Gap

- Enhancing the security of interdependent systems depends critically on **human decisions to allocate limited security resources**
- Numerous works have been proposed for improving the security of interdependent systems (e.g., [Laszka et al. 2015])
 - They are based on traditional models that defenders make **completely logical security decisions** and evaluate risks in **a rational manner**
- Nonetheless, research in **behavioral economics** indicates that humans are only **partially rational** and often deviate from the traditional models
 - This **motivates** the need to **guide human defenders** towards better allocation

Key Message of This Thesis

“By integrating, graph theoretic approaches based, node ranking, and learning based methods, we can:

- 1. Develop intelligent, adaptive, and data-driven cyber defense frameworks using graph representations, ranking, clustering, and meta-learning to reduce security cost and improve resilience. , &*
- 2. Introduce stress-testing frameworks to evaluate the robustness, resilience, and failure modes of cyber attack-graph defense strategies under adversarial attacks.”*

Thesis Milestone

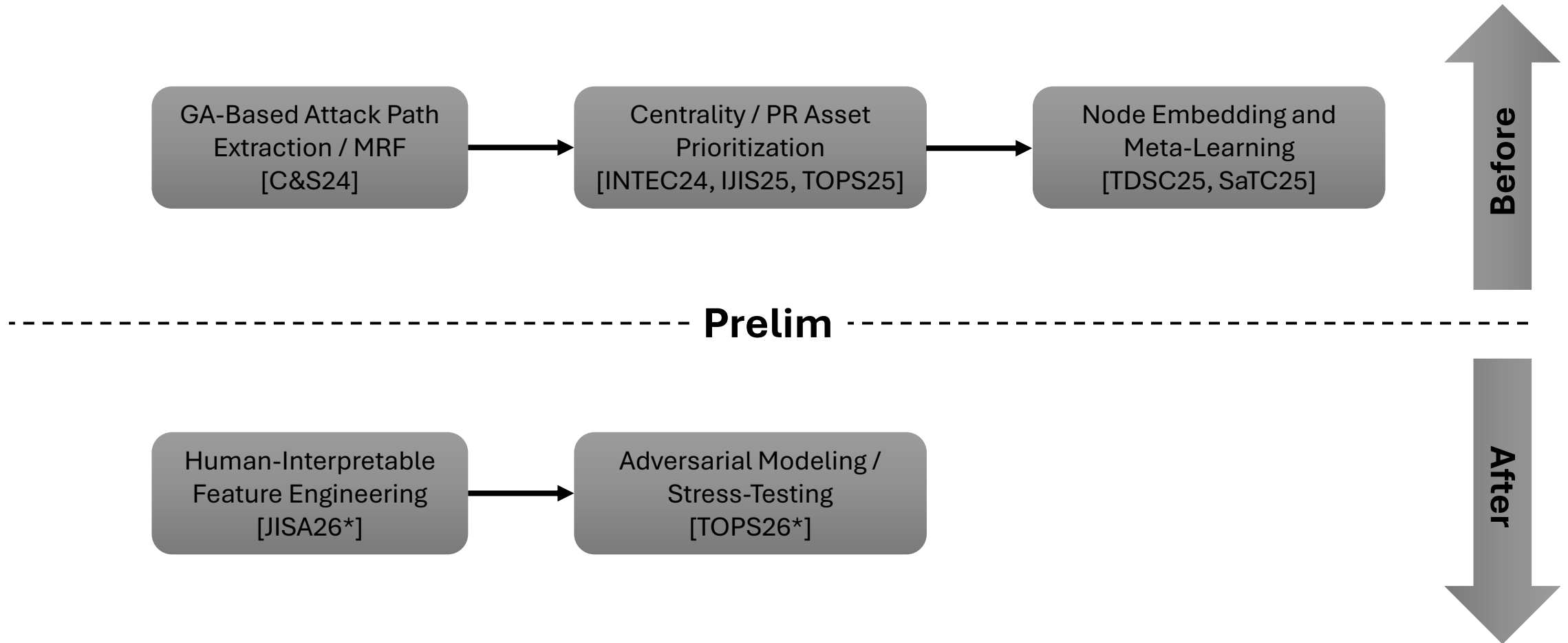
- **Before Preliminary Exam (August / 2023 – December / 2025):**

- Developing a genetic-based approach to learn and predict the most probable attack paths that adversaries may exploit to breach the system [C&S24]
- Investigating using node-ranking algorithms to guide a rational defender to allocate security resources based on asset importance [IJIS25, TOPS25, INTCEC24]
- Designing a meta-learning framework that forecasts and recommends optimal or near-optimal resource allocation strategies for unseen systems [TDSC25, SaTC25]

- **After Preliminary Exam (December / 2025 – May / 2026):**

- Developing a hand-crafted attack graph feature extraction method representing the system using human-interpretable attributes rather than latent embeddings [JISA26*]
- Analyzing the performance of different resource allocation methods under adversarial attacks, where attackers intentionally perturb the network topology [TOPS26*]

Thesis Timeline



Research Contribution

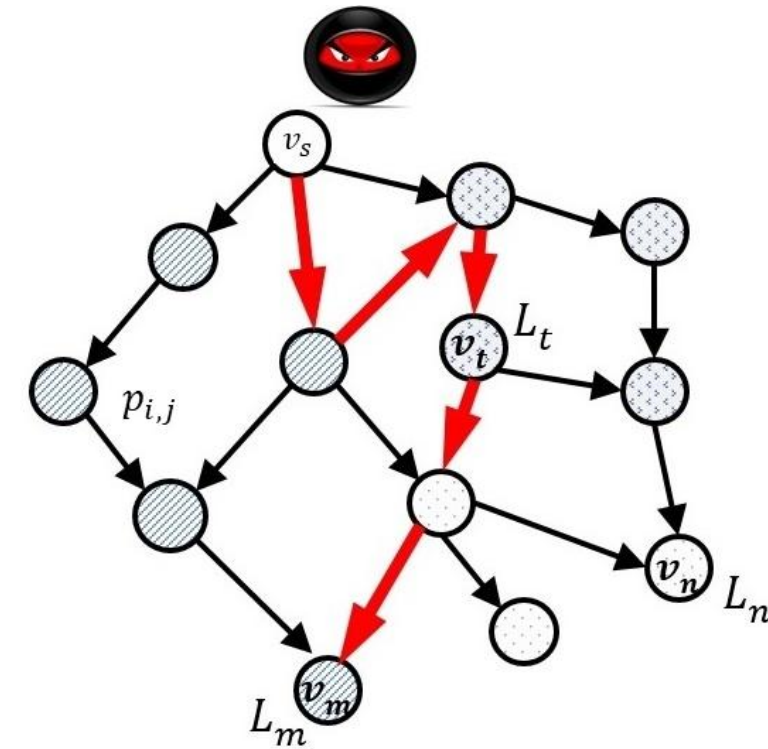
- Proposing **novel resource allocation** for **securing** interdependent systems
- Utilizing an evolutionary **Genetic Algorithm (GA)** for **potential** attack path **estimation** based on **a novel multi-objective** function
- Utilizing fundamental graph-theoretic, and **linkage analysis** approaches to allocate the limited security resources and **prioritize significant** assets
- Constructing meta-learning (RWs & **Hand-Crafted Features**) dataset, training ML classifiers to predict **near** / **optimal** resource allocation method
- Allocating resources based on the **severity** of node **communities** formed from nodes, **sharing** similar **latent features**
- Stress-testing **state-of-the-art** resource allocation methods under varied set of **adversarial attacks**

Overview of Interdependent Systems Framework

Threat Model of the System

- The system is modeled using attack graph $G = (V, E)$
- Nodes (V) represent **entries**, intermediates and **assets**.
- Edges (E) represent attack paths.
- Direct arc $(v_i, v_j) \in E$ is occurred when v_i is breached and then attacking v_j
- The weight $x_{i,j} \in (0, 1]$ represents the initial investment.
- The probability $p_{i,j}(x_{i,j}) = \exp(-x_{i,j})$ indicates the chance of an attacker compromising node v_j from v_i .
- The probability of a successful attack path P is:

$$\prod_{(v_i, v_j) \in P} \exp(-x_{i,j})$$

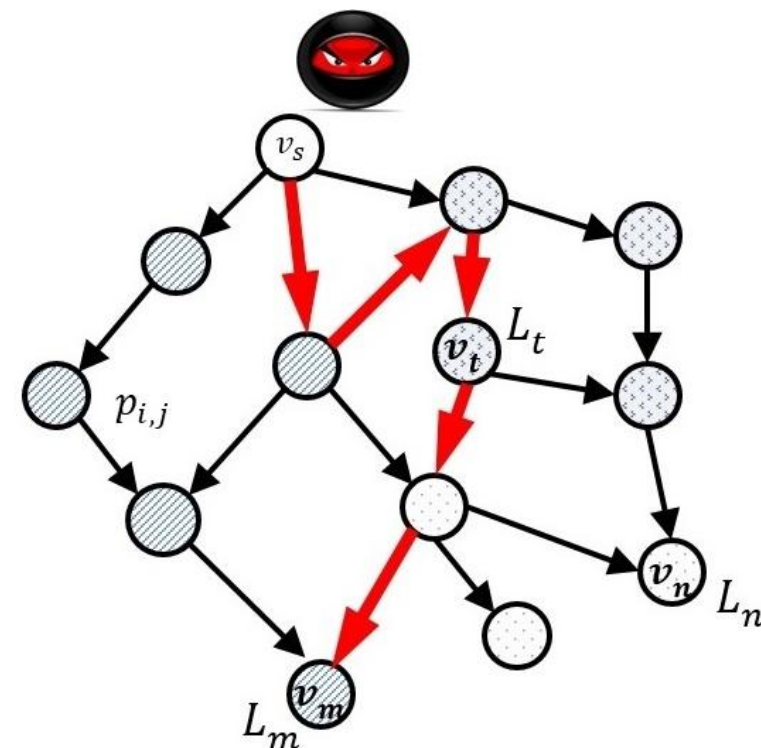


Directed Acyclic Attack Graph (DAG)

Defense Model of the System

- The defender **D** is responsible for protecting a set of assets $V_m \subseteq V$
- If a critical asset $v_m \in V_m$ is compromised, the defender incurs a financial loss $L_m \in R > 0$.
- L_m represents the system's financial loss upon the breach of $v_m \in V_m$
- Resources are distributed on edges $(v_i, v_j) \in E$, represented by $x_{i,j}$
- Defender **D** tends to allocate her investments to minimize this cost function:

$$cost = \sum_{v_m \in P} L_m * \left(\max_{P \in P_m} \prod_{(v_i, v_j) \in P} \exp(-x_{i,j}) \right)$$



Directed Acyclic Attack Graph (DAG)

Research Allocation Performance Metric

- Cost Relative Reduction (CR %): Measure the relative reduction in expected security cost **before** and **after** applying a **defense allocation** strategy. CR % is given by:

$$CR(P) = \frac{|C(P) - \hat{C}(P)|}{C(P)} * 100 \%$$

- Note that $C(P)$ and $\hat{C}(P)$ are the expected cost before and after the resource allocation respectively.
 - Before the allocation $C(P)$ is given by :
 - After the allocation $\hat{C}(P)$ is given by :

$$\sum_{v_m \in P} L_m * \prod_{(v_i, v_j) \in P} \exp(-x_{i,j}^0)$$

$$\sum_{v_m \in P} L_m * \prod_{(v_i, v_j) \in P} \exp(-x_{i,j})$$

Real Interdependent Systems Modeled as Attack Graphs

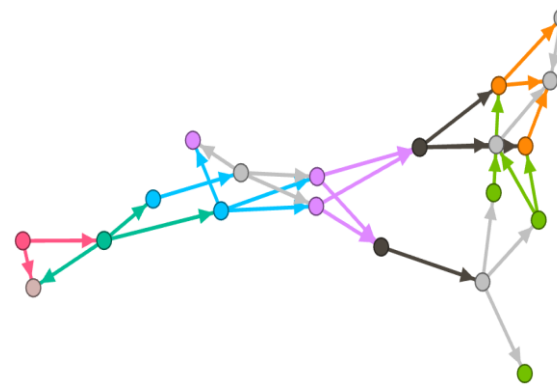
- **Gephi** visualizations of the four real-world CPS attack graphs: DER, SCADA, E-commerce, and VOIP



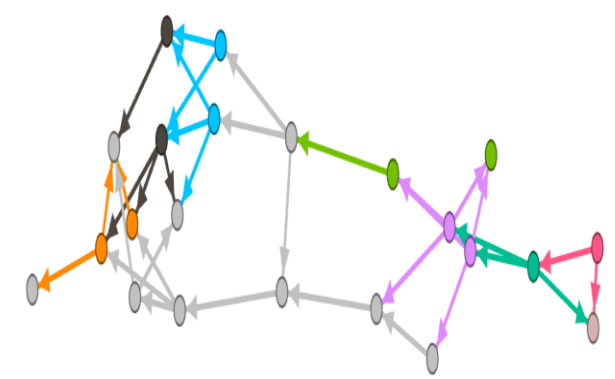
DER



SCADA



E-commerce



VOIP

Real Interdependent Systems Modeled as Attack Graphs

- These systems are different since they have different **topological** characteristics: **#of nodes**, **#of edges**, **#of assets**, and many others
- The input of these systems is **modeled** using **an edge list** in addition to **a vector of asset losses**

1	3
1	5
1	6
1	8
1	9
.	
.	
.	
1	12
8	13
10	13
11	12
12	13
Loss	0 63 39 0 0 0 33 0 0 22 0 90 45

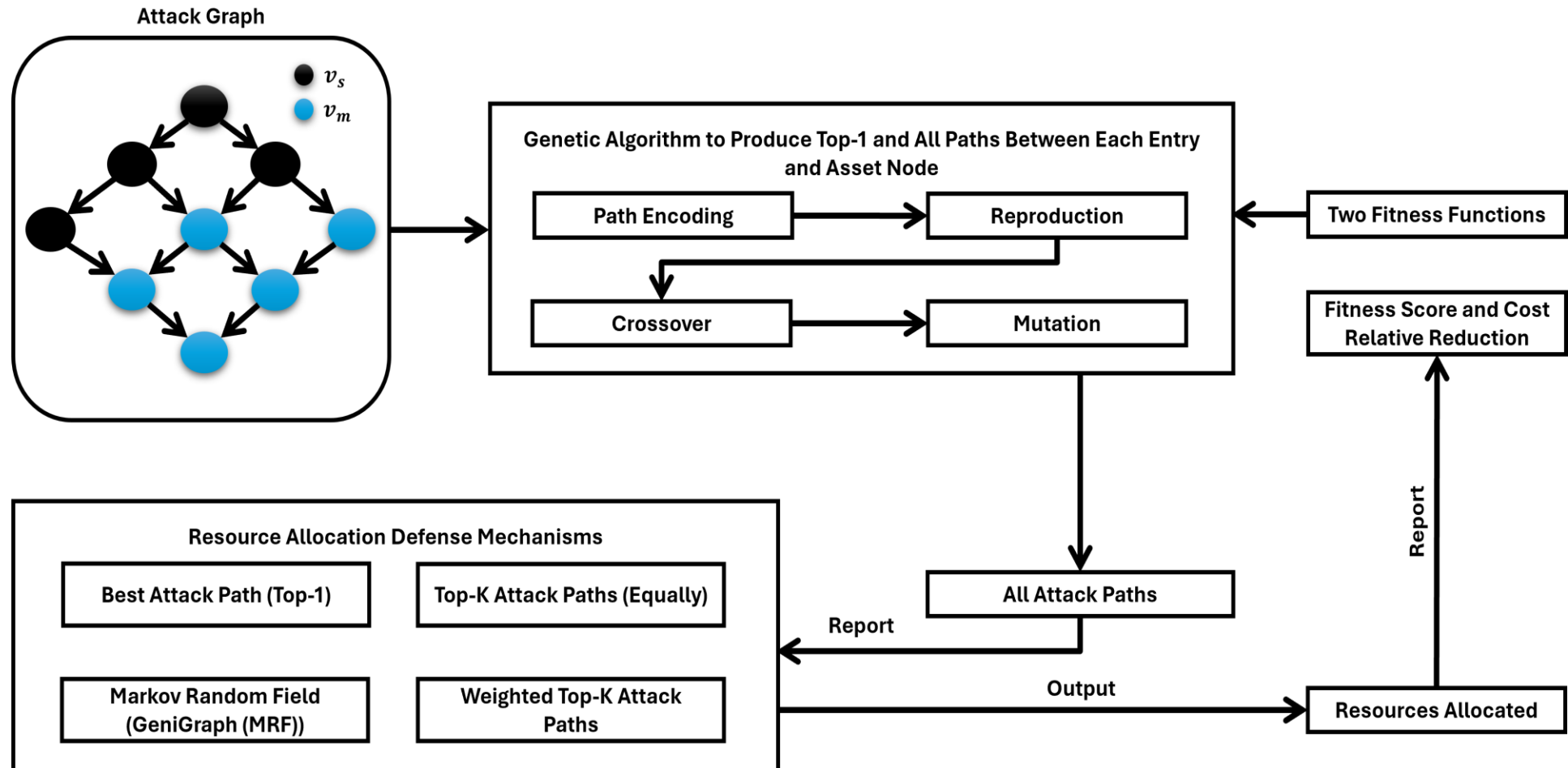


SCADA

GeniGraph: Genetic-Based Defense Resource Allocation

Publication: *Mohammad Ryiad Al-Eiadeh and Mustafa Abdallah, “Genigraph: A Genetic-Based Novel Security Defense Resource Allocation Method for Interdependent Systems Modeled by Attack Graphs”, **Computers & Security**, 2024.*

Overview Diagram of GeniGraph



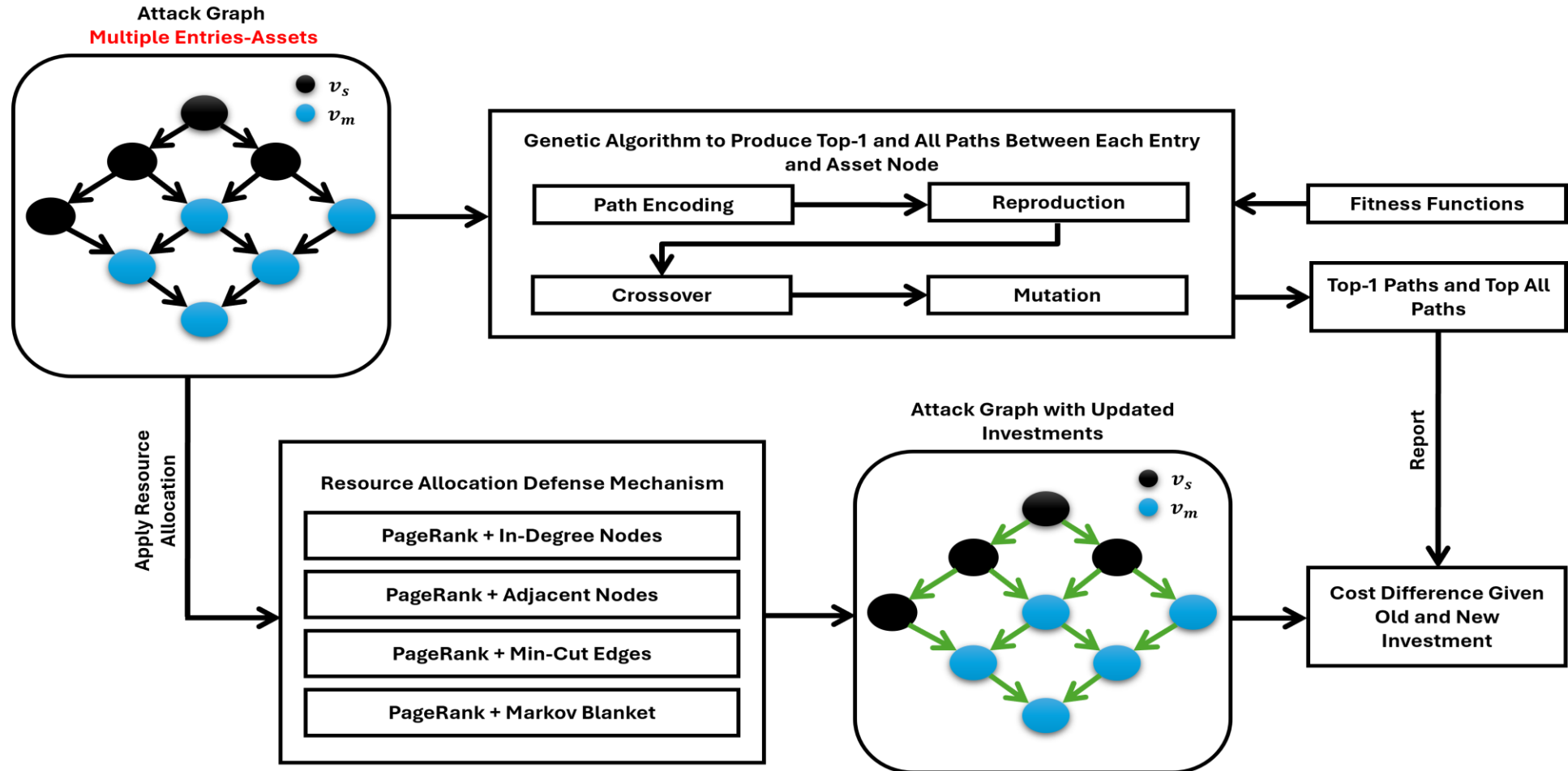
Summary of Contributions

- We developed a **GA-guided decision-maker** to infer likely attack paths, including **top-1**, **top-K**, and **all potential attack paths**
- **GeniGraph(MRF)** protects edges **surrounding critical assets** and remains robust even when attackers' **entry points or numbers change**
- GeniGraph **outperformed** baseline approaches across most experiments, including different attack scenarios and varying **initial** investment settings

PR-DRA: PageRank-based defense resource allocation methods for securing interdependent systems modeled by attack graphs

Publication: *Mohammad Ryiad Al-Eiadeh and Mustafa Abdallah, “PR-DRA: PageRank-based Defense Resource Allocation Methods for Securing Interdependent Systems Modeled by Attack Graphs,” **International Journal of Information Security**, 2025.*

Overview Diagram of PR-DRA



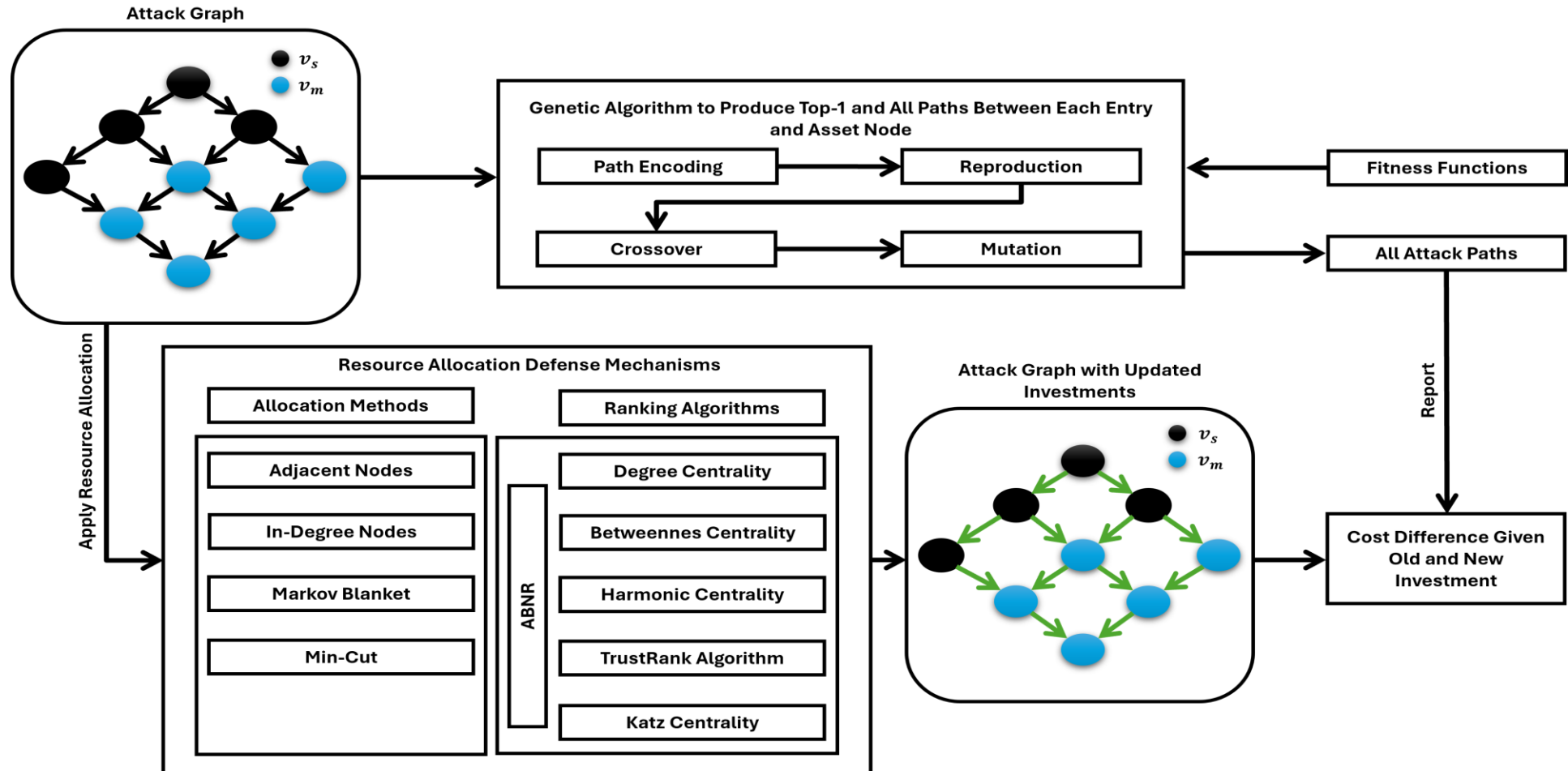
Summary of Contributions

- We **integrated** four defense allocation mechanisms to guide a single defender for securing interconnected systems
- We **utilized** PageRank-based centrality (PR) to rank critical assets based on weighted incoming links
- We **used** rankings to proportionally allocate security budgets
- We **showed** that the proposed methods have **low sensitivity** to attack entry points (except Min-Cut).

CBDRA-IS: Centrality-Based Defense Resource Allocation for Securing Interdependent Systems

Publication: *Mohammad Ryiad Al-Eiadeh and Mustafa Abdallah, “CBDRA-IS: Centrality-Based Defense Resource Allocation for Securing Interdependent Systems,” **ACM Transactions on Privacy and Security (TOPS)**, 2025.*

Overview Diagram of CBDRA-IS



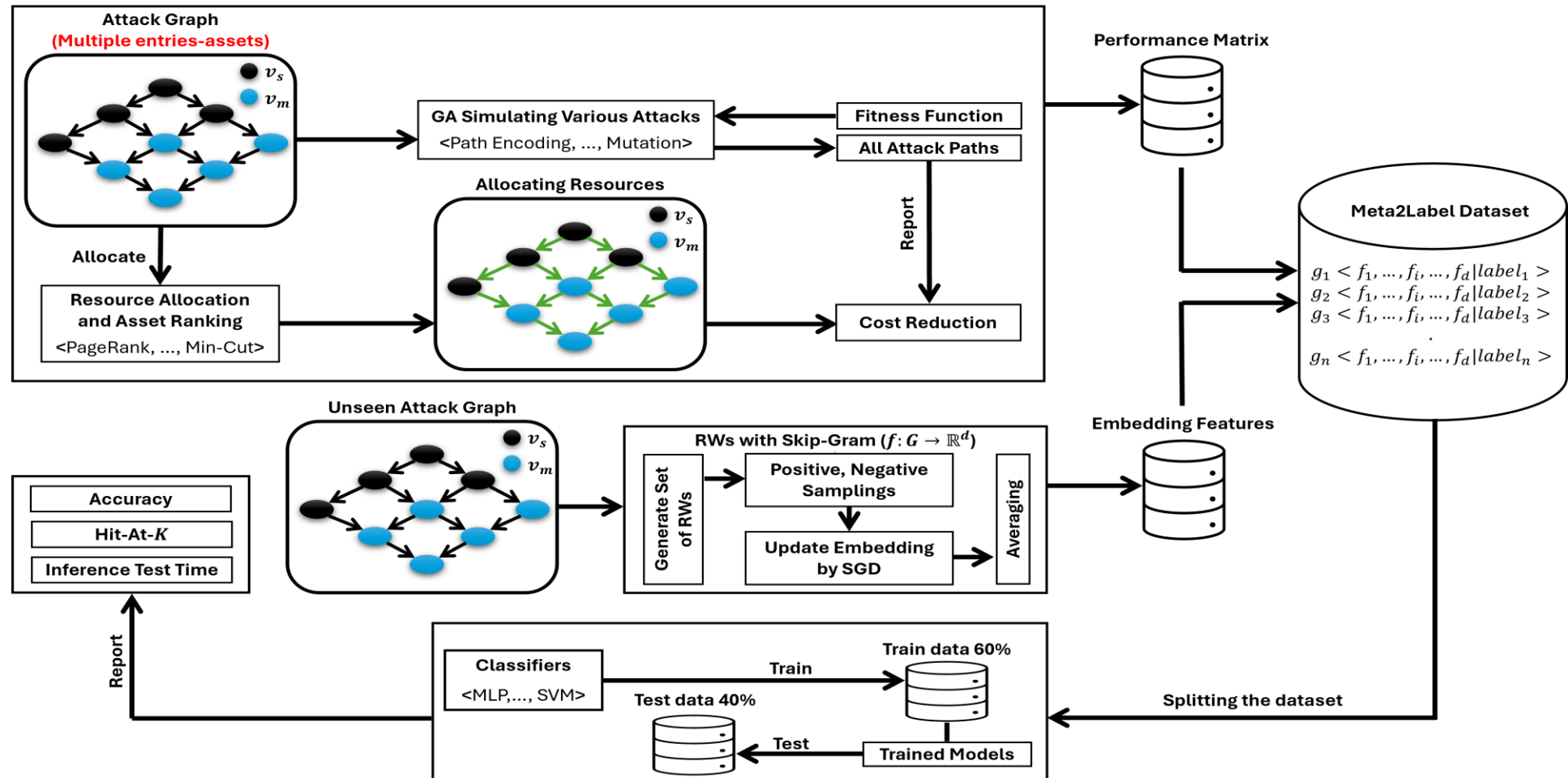
Summary of Contributions

- We ranked assets using five **centrality-based algorithms**, each capturing different **structural characteristics**
- We **integrated** graph-theoretic defense strategies and node-ranking algorithms to produce an efficient resource allocation model
- We showed that CBDRA-IS **achieved** significant cost reduction under **various attack types** such as cascading failures
- Our assessments **proved** that CBDRA-IS is **stable** even with different investments distribution, time constraints, and behavioral biases

Auto-Sec: Meta-Learning for Automated Efficient Security Resource Allocation on Attack Graphs

Publication: *Mohammad Ryiad Al-Eiadeh and Mustafa Abdallah, “Auto-Sec: Meta-Learning for Automated Efficient Security Resource Allocation on Attack Graphs,” IEEE Transactions on Dependable and Secure Computing (TDSC), 2025.*

Overview Diagram of Auto-Sec



Main Idea of Meta-Learning

- The use of past **learned patterns** of seen attack graphs to predict the **best resource allocation strategy** for new, unseen graphs
- Minimize the **gap** between the **actual optimal strategy** and the meta-learner's **predicted** performance in terms of cost reduction.
- In other words, we teach the meta-learner to **predict the optimal or near-optimal strategy** for unseen graphs without evaluating all strategies.

Main Idea of Meta-Learning (cont.)

- We **aim** to **minimize**:

$$\sum_{i=1}^{NG} \left(\sum_{j=1}^{NR} CR \left(G_i(R_j) \right) - CR \left(\phi(\psi_i, R_j, \Theta) \right) \right)$$

where:

- NG : the number of attack graphs
- NR : the number of resource allocation
- R_j : the j th resource allocation strategy
- $\phi(\cdot)$: the meta-learner prediction model
- Θ : the predicted model parameter

Performance Assessment for Unseen Systems

- For a **new attack graph** G_{new} , we learn the embedding and then feed it to the trained meta-learner for **inferring** or **predicting** the resource allocation
- Meta-Learner **Inference Rule**:

$$R_j^* \in \arg \max_{R_j \in \Omega} CR \left(\phi \left(\psi(G_{new}), R_j(G_{new}), \Theta \right) \right)$$

where:

- R_j^* : the optimal or near optimal resource allocation
- Ω : the set of the twenty-seven-resource allocation strategies

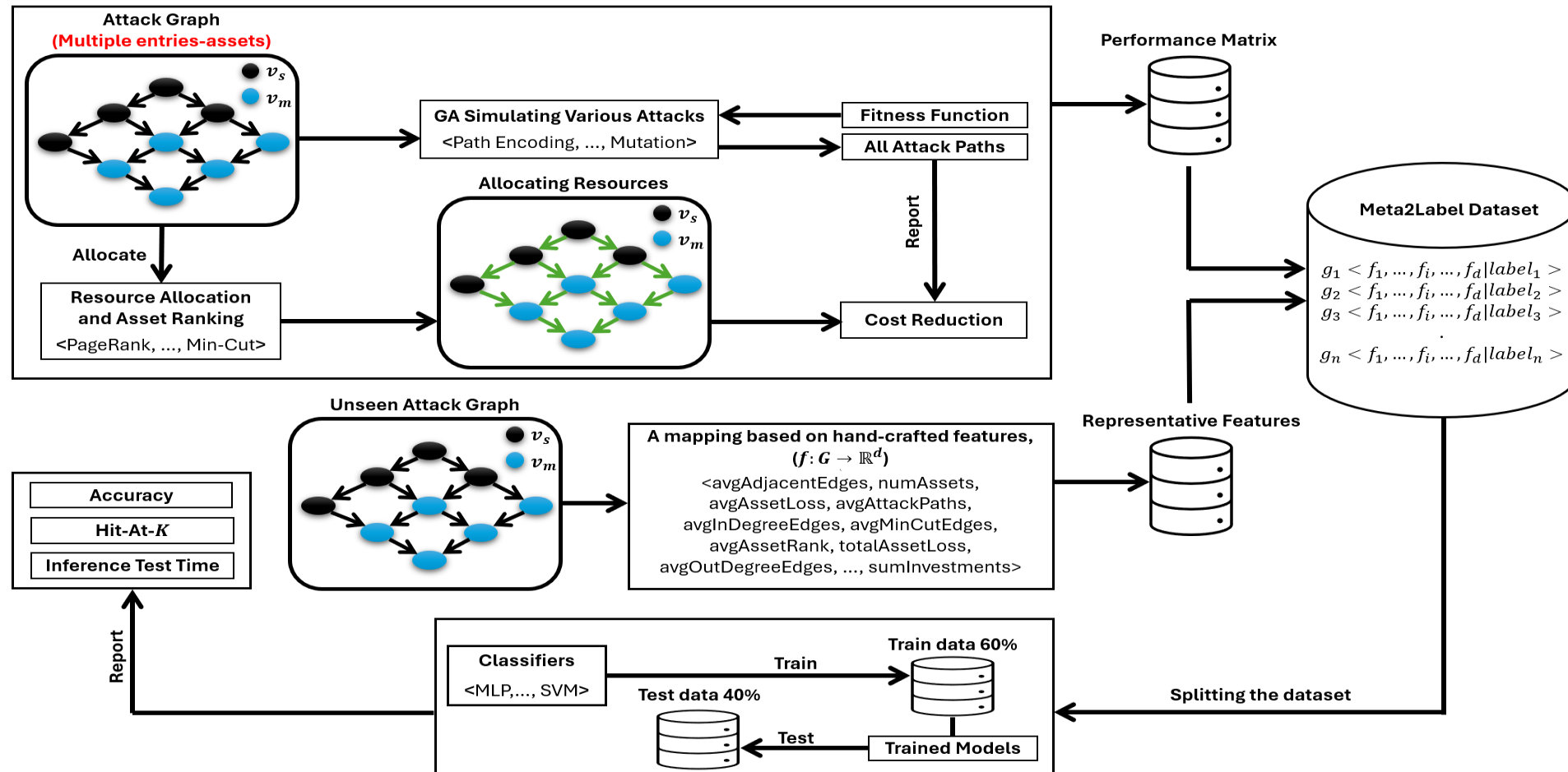
Summary of Contributions

- We **developed** Auto-Sec, a **meta-learning** framework that automatically **predicts** the **optimal** or **near-optimal** security allocation strategy
- We **used RWs-based embeddings** and feature selection to extract meaningful latent graph features for **accurate** and **fast** prediction
- Auto-Sec **outperformed** baselines in accuracy, speed, and cost reduction, showing that Auto-Sec framework is robust and scalable for securing interdependent systems

Auto-Sec-HS: Meta-Learning Approach Based on Hand-Crafted Domain-Based Features for Efficient Security Resource Allocation on Attack Graphs

Publication: *Mohammad Ryiad Al-Eiadeh and Mustafa Abdallah, “Auto-Sec-HS: Meta-Learning Approach Based on Hand-Crafted Domain-Based Features for Efficient Security Resource Allocation on Attack Graphs,” Journal of Information Security and Applications (JISA), 2026. (Under Review)**

Overview Diagram of Auto-Sec-HS



Main Research Contributions

- We propose **a meta-learner** for predicting efficient resource allocation strategies using attack-graph domain-knowledge features
- We construct **a meta-feature** dataset using an **engineered hand-crafted graph features** and allocation **performance matrix**
- We **validate** the framework on diverse real-world datasets, where it **outperforms** baselines and **GNN-based** methods

Motivation: Hand-Crafted vs. Latent Space

- The **latent graph embeddings** capture **complex** structures but are **difficult** to **interpret** and **explain**
- Node Embeddings like **RWs** and **Skip-Gram** generate **opaque black-box** representations
- Limited **explainability** reduces **understanding** and **trust** of security **decision-making**
- **Hand-crafted** attack-graph features improve **interpretability** and support **explainable** defense allocation; vary important for FS

Hand-Crafted Features Extracted from Attack Graphs

- We **designed 23** domain-driven hand-crafted features capturing **structural**, **severity**, asset **importance**, and **local/global connectivity** properties to generalizable attack-graph **representation**

Feature	Notation	Description
numNodes	$ V $	Total number of nodes in the attack graph
numEdges	$ E $	Total number of directed attack transitions
numEntries	$ V_s $	Number of entry nodes
numAssets	$ V_m $	Number of critical assets
sumInvestments	Σx_{ij}	Total security investment
avgInvestments	$(1/ E) \Sigma x_{ij}$	Average investment per edge
graphSeverity	$\psi(G, L, x)$	Overall system severity
totalAssetLoss	ΣL_m	Total asset loss
avgAssetLoss	$(1/ V_m) \Sigma L_m$	Average asset loss
sumAssetRanks	Σr_m	Total asset importance score
avgAssetRank	$(1/ V_m) \Sigma r_m$	Average asset importance

Hand-Crafted Features Extracted from Attack Graphs (cont.)

totalInDegreeEdges	$\Sigma \deg^-(v_m)$	Total incoming asset edges
avgInDegreeEdges	$(1/ V_m) \Sigma \deg^-(v_m)$	Average incoming edges per asset
totalOutDegreeEdges	$\Sigma \deg^+(v_m)$	Total outgoing asset edges
avgOutDegreeEdges	$(1/ V_m) \Sigma \deg^+(v_m)$	Average outgoing edges per asset
avgAdjacentEdges	$(1/ V_m) \Sigma (\deg^-(v_m) + \deg^+(v_m))$	Average adjacent edges per asset
totalMB	$\Sigma MB(v_m) $	Total Markov Blanket connections
avgMB	$(1/ V_m) \Sigma MB(v_m) $	Average Markov Blanket size
totalMinCutEdges	$\Sigma \lambda(v_s, v_m)$	Total min-cut edges
avgMinCutEdges	$(1/ V_m) \Sigma \lambda(v_s, v_m)$	Average min-cut edges per asset
frobeniusNorm	$\ A\ _F$	Frobenius norm of the adjacency matrix
totalAttackPaths	$\Sigma P(v_s, v_m) $	Total number of attack paths
avgAttackPaths	$(1/ V_m) \Sigma P(v_s, v_m) $	Average attack paths per asset

Evaluation of Auto-Sec-HS

- Same **real-world testbed** used to evaluate **Auto-Sec**
- Same **hyper-parameter** settings; **GA**, **Centralities**, and **Learners**
- Same **baseline** for comparison; **ISAC**, **GLOBAL best**, and **ARGOSMART**
- Same **classifiers** employed; **CART**, **FM**, **SVM**, ..., and **LR**
- Same **evaluation** metrics used; **Accuracy (Hit @ 1)**, **Hit @ {5, 10}**, and **Cost Relative Reduction**

Evaluation—Auto-Sec-HS vs. Baselines

Algorithm / Metrics	Accuracy	Hit-at-5	Hit-at-10	Avg CR (%)
LeNet-5	0.25	0.65	0.9	40.87
MLP	0.2	0.66	0.88	39.65
FM	0.13	0.63	0.9	40.64
SVM	0.15	0.38	0.85	37.23
LR	0.18	0.75	0.93	40.61
KNN	0.2	0.7	0.83	40.26
CART	0.23	0.85	0.98	42.38
LibLinear	0.23	0.75	0.9	41.26
XGBoost	0.23	0.83	0.95	42.57*
M-NB	0.28	0.73	0.85	39.49
ARGOSMART	0.18	0.63	0.83	40.1
Global Best	0.2	0.45	0.88	37.7
ISAC-Kmeans	0.2	0.75	1	41.68
ISAC-Kmeans++	0.29	0.73	1	41.39
ISAC-HDBSCAN	0.2	0.45	0.88	37.7

- **XGBoost & CART** depict the **best performance** at CR (**42.10 %**, **42.38 %**) Respectively.

Evaluation—Auto-Sec-HS Under Feature Selection

- We used **mRMR filter** to select the **top-15 meaningful** features

Algorithm / Metrics	Acc	Hit-at-5	Hit-at-10	Avg CR (%)
Cart	0.2	0.8	0.93	41.08*
XGBoost	0.23	0.68	0.93	42.01
ARGOSMART	0.18	0.7	0.88	40.72
Global Best	0.2	0.45	0.88	37.7
ISAC-Kmeans	0.23	0.6	0.95	41.02
ISAC-Kmeans++	0.2	0.53	0.95	40.64
ISAC-HDBSCAN	0.2	0.45	0.88	37.7

Category	Features
Selected Features	avgAdjacentEdges, numAssets, avgAssetLoss, avgAttackPaths, avgInDegreeEdges, avgMinCutEdges, avgAssetRank, totalAssetLoss, avgOutDegreeEdges, numEntries, avgMB, avgInvestments, totalMB, totalMinCutEdges, graphSeverity
Dropped Features	sumAssetRanks, numEdges, totalOutDegreeEdges, numNodes, totalInDegreeEdges, frobeniusNorm, totalAttackPaths, sumInvestments

- XGBoost & CART** depict the **best performance** at CR (**42.01 %**, **41.08 %**) Respectively.

Evaluation—Auto-Sec-HS Against GNN-Like Approaches

Algorithm	Hand-Craft Features	GNN	GAT	GraphSage	ChebNet
			Hit-at-5		
CART	0.85*	0.55	0.7	0.5	0.68
XGBoost	0.83	0.58	0.63	0.68	0.68
			Hit-at-10		
CART	0.98*	0.75	0.83	0.75	0.9
XGBoost	0.95	0.9	0.88	0.93	0.9
			Avg CR %		
CART	42.38	35.97	40.27	37.13	38.81
XGBoost	42.57*	39.32	15.79	39.51	39.1

- **XGBoost & CART** depict the **best performance** according to **Hit @ {5,10}** and **CR**

Summary of Contributions

- We **developed 23 hand-crafted attack graph features** for robust, **explainable**, and **accurate** prediction
- Auto-Sec-HS **outperformed** baselines in **cost reduction**, showing that the framework is **reasonable** for securing interdependent systems

RA-AGSRA: A Robustness Analysis of Attack-Graph-Driven Security Resource Allocation Schemes for Interdependent Systems under Graph-based Adversarial Attacks

Publication: *Mohammad Ryiad Al-Eiadeh and Mustafa Abdallah, “A Robustness Analysis of Attack-Graph-Driven Security Resource Allocation Schemes for Interdependent Systems under Graph-based Adversarial Attacks,” **ACM Transactions on Privacy and Security (TOPS)**, 2026. (Under Review)**

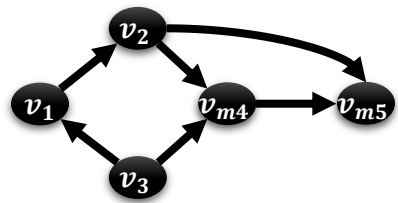
Main Research Contributions

- Introduce **a stress-testing framework** for evaluating the robustness of **attack-graph-based** defense strategies against **adversarial attacks**
- Model **realistic adversarial manipulations**, including **asset removal**, **phantom asset insertion**, and **asset-label flipping**
- Evaluate the robustness and **failure modes** of **PR**, **In-Degree**, **MB**, and **Min-Cut** defense strategies under **adversarial attack-based** scenarios
- Introduce **reasonable-performance ratios** measurement to **inform** robustness analyses of such defense methods
- Provide insights into why graph-driven defense strategies **fail** under **adversarial manipulation**

Modeling Adversarial Attacks

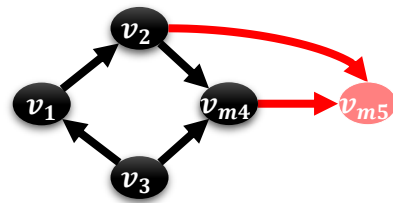
- Adversarial attacks **manipulate** attack graphs to **mislead** graph-analysis models and **decision-makers**, **reducing** the effectiveness of security resource allocation in **CPS**
- We **analyze** the proposed resource allocation methods under **four** white-box **adversarial attacks** that **manipulate** attack-graph structures to mislead the **defender** decision
- Here, we **shift** the **focus** from **proposing** defense strategies to **critically** evaluating and exposing the **failure modes** of **state-of-the-art** approaches under **adversarial attacks**

Modeling Adversarial Attacks (cont.)



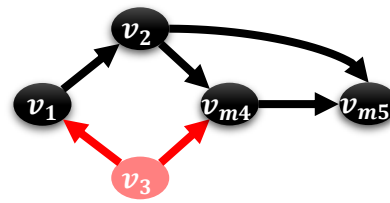
0	1	0	0	0
0	0	0	1	1
1	0	0	1	0
0	0	0	0	1
0	0	0	0	0

Original Matrix A



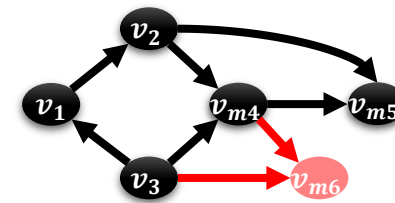
0	1	0	0	0
0	0	0	1	0
1	0	0	1	0
0	0	0	0	0
0	0	0	0	0

Perturbed Matrix $\tilde{A}$
Remove Asset Attack



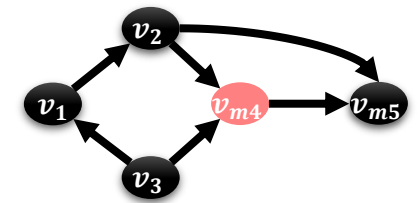
0	1	0	0	0
0	0	0	1	1
0	0	0	0	0
0	0	0	0	1
0	0	0	0	0

Perturbed Matrix $\tilde{A}$
Remove non-Asset Attack



0	1	0	0	0	0
0	0	0	1	1	0
1	0	0	1	0	1
0	0	0	0	1	1
0	0	0	0	0	0
0	0	0	0	0	0

Perturbed Matrix $\tilde{A}$
Add Phantom Attack



0	1	0	0	0
0	0	0	1	1
1	0	0	1	0
0	0	0	0	1
0	0	0	0	0

Keeps Original Matrix A
Flip Asset Attack

Modeling Adversarial Attacks (cont.)

Attack Model	Core Manipulation	Impact on Defender
Remove Assets	Hide critical asset nodes and their connected edges	Breaks dependency chains and conceals critical attack paths, leading to under-protection of important assets
Remove Non-Assets	Hide intermediate structural nodes and their links	Distorts graph topology, node ranking, and critical-zone identification, misleading prioritization strategies
Add Phantom Assets	Inject fake assets and artificial connections	Expands the attack surface and misguides defenders toward protecting non-existent targets
Flip Asset Flags	Change asset labels to appear as normal intermediate nodes	Excludes critical assets from ranking and allocation processes without altering graph structure

- Key: Adversaries **intently distort** the attack graph **topology**, causing graph-based defense strategies to **misidentify assets**, **underestimate** risks, and allocate security resources **inefficiently**

Evaluation of Auto-Sec-HS

- Same 10-real world datasets, parameter settings used in **CBDRA-IS**
- We introduce Δ_{CR} to **measure** the impact of **adversarial perturbations** on **cost reduction** by comparing **defense allocations** from the **original** and the **perturbed** attack graphs

$$\Delta_{CR} = \frac{|C(P) - \hat{C}_G(P)| - |C(P) - \hat{C}_{\tilde{G}}(\tilde{P})|}{C(P)} * 100 \%$$

where:

- G : the original attack graph
- $\tilde{G}$: the perturbed attack graph
- $\tilde{P}$: the learned path from $\tilde{G}$

Evaluation of Auto-Sec-HS (cont.)

- We introduce the **Reasonable Performance Ratio (RPR)** as the total number of **reasonable outcomes** under **adversarial attacks divided** by the total number of test cases (i.e., attack graphs)

$$RPR = \frac{1}{N} \sum_{i=1}^N \varphi(\Delta_{CR}^i \leq \varepsilon_0)$$

$$\varphi(\cdot) = \begin{cases} 1, & \Delta_{CR} \leq \varepsilon_0 \\ 0, & O.W. \end{cases}, \text{ for every } i_{th} \text{ attack graph}$$

where:

- N : the number of attack graphs
- ε_0 : the reasonable threshold (set to 15 %)

Evaluation of Auto-Sec-HS (cont.)

- Performance **assessment** of state-of-the-art without **launching** adversarial attacks in terms of cost relative reduction and **mean rank**

Graph ID	Risk Based Defense	Defense In Depth	Behavioral	Min-Cut	PR + In-Degree	PR + Adjacent	PR + MB	PR + Min-Cut
DER	48.1989	46.4739	46.8254	49.9316	81.4969	72.9254	61.3205	50.5632
HG2	57.976	56.5402	57.3634	67.4211	69.1804	58.5155	54.0218	60.6597
E-commerce	53.2442	46.4739	51.9271	49.3837	79.2421	79.2421	70.2237	36.949
HG1	89.4981	86.4665	89.4846	90.5572	94.2859	94.455	84.7384	94.6209
ABSNP	12.5502	12.611	12.5493	29.1888	35.6655	34.7906	25.5073	30.5459
ASS2009	6.1708	4.9931	6.0507	23.8544	16.3492	15.7127	11.925	18.821
VoIP	52.4664	43.5282	51.3154	53.0712	91.4601	91.4601	78.8511	45.8593
AWS03	9.8434	9.0205	9.6597	30.3405	18.664	16.6087	13.1528	21.9349
ASFS3	4.7342	4.6295	4.7034	12.5715	10.641	7.9211	5.9226	11.9699
SCADA	69.3157	63.2121	70.1496	71.6077	69.8669	58.4767	54.0027	79.6113
Mean Rank	3.5	1.7	2.8	6	6.8*	5.8	3.7	5.7

- PR combinations depict the best approaches, especially **PR + In-Deg**

Evaluation of Auto-Sec-HS (cont.)

- Application of **adversarial perturbations** to attack graphs, including removed assets, fake assets, removed non-assets, and flipped asset labels

Graph ID	Removed Assets	Number of Phantoms	Removed non-Assets	Flipped Assets' Label
DER	{4, 20}	2	{15}	{14}
HG2	{7, 9}	2	{15}	{5}
E-commerce	{20}	1	{10}	{7}
HG1	{5}	1	{4}	NaN
ABSNP	{17, 11}	2	{5}	{2}
ASS2009	{19, 26}	2	{10}	{31}
VoIP	{18}	1	{9}	{3}
AWS03	{39, 25}	3	{1}	{11}
ASFS3	{3, 8}	1	{5}	{13}
SCADA	{13}	2	{8}	{7}

- Coalition (or **collaborative**) attackers work together to breach the systems

Evaluation of Auto-Sec-HS (cont.)

- Performance **assessment** of state-of-the-art after **launching** adversarial attacks in terms of cost relative reduction and **mean rank**, and **RPR**

Graph ID	Risk-Based	Defense In Depth	Behavioral	Min-Cut CR (%)	PR + In-Deg	PR + Adjacent	PR + MB	PR + Min-Cut
DER	2.0139	7.5692	0.6669	30.9263	26.1952	22.3512	15.6844	28.1059
	.							
	.							
SCADA	48.7927	42.9537	48.0836	73.1675	61.9673	52.7378	48.8169	72.1921
Mean Rank	3.8	4	3.8	5.9*	4.6	4.4	4.4	5.1
				ΔCR				
DER	46.185	38.9047	46.1585	19.0053	55.3017	50.5742	45.6361	22.4573
	.							
	.							
SCADA	20.523	20.2584	22.066	-1.5598	7.8996	5.7389	5.1858	7.4192
RPR	0.4	0.4	0.4	0.4	0.4	0.5*	0.5*	0.4

- All methods, **especially ranking-based** approaches, **illustrate significant performance degradation** under **adversarial attacks**

Frobenius Norm Analysis on Attack-Surface Manipulation

- Phantom node **introduces** new **nonzero** entries in the adjacency matrix A
 - Which **inflates** the **perceived attack surface**
- The **increase** in **connectivity mass** potentially **misleading defenders** toward deceptive or **low-value** structures

$$\| \tilde{A} \|_F = \| A \|_F + \delta, \text{ for some } \delta > 0 \text{ implies } \| \tilde{A} \|_F > \| A \|_F$$

- Node or edge removals **eliminate connectivity** information from A
 - Which **shrinks** the **perceived attack surface**
- The **resulting reduction** in graph **connectivity** potentially **hiding critical** attack paths and **degrading** defense allocation **decisions**

$$\| \tilde{A} \|_F = \| A \|_F - \delta', \text{ for some } \delta' > 0 \text{ implies } \| \tilde{A} \|_F < \| A \|_F$$

Summary of Contributions

- Adversarial **manipulation** of attack graphs **significantly degrades** graph-based defense allocation **strategies**, especially under **combined** attacks
- Defense performance **strongly** depends on graph **structures**, making ranking-based approaches **sensitive** to graph **perturbations**
- Attack-surface **inflation** and **deflation** alter **perceived** graph connectivity, **motivating resilient** and **adversary-aware defense** frameworks

Conclusion

Conclusion

- We developed **graph-theoretic**, **asset Prioritization** and **optimization-driven** frameworks for **intelligent** security resource allocation
- We introduced **a meta-learning framework** (based on **latent** and **hand-crafted**) framework that **predicts** effective defense strategies for **unseen** interdependent systems
- We established **adversarial stress-testing** framework to expose the **failure modes** of graph-driven defense strategies under **adversarial** attacks

List of Publications

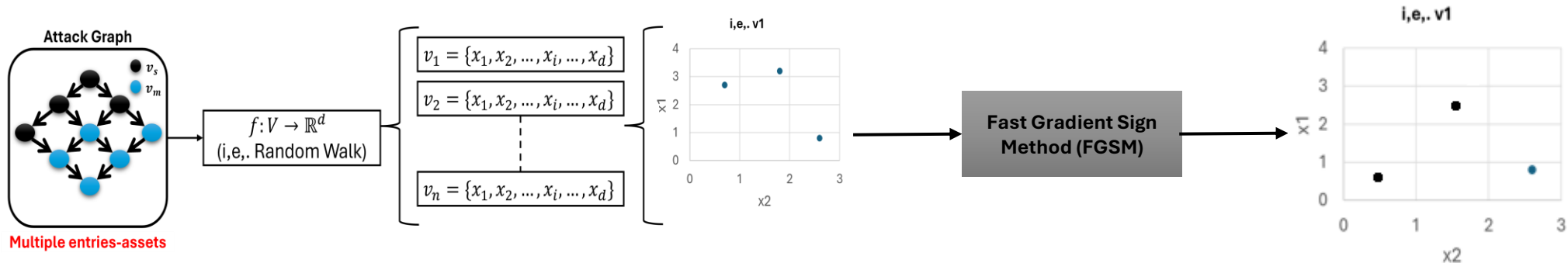
Thesis Publications

- **Mohammad Ryiad Al-Eiadeh** and Mustafa Abdallah, “A Robustness Analysis of Attack-Graph-Driven Security Resource Allocation Schemes for Interdependent Systems under Graph-Based Adversarial Attacks”, *ACM Transactions on Privacy and Security (TOPS)*, 2026. **Under Review***
- **Mohammad Ryiad Al-Eiadeh** and Mustafa Abdallah, “Auto-Sec-HS: Meta-Learning Approach Based on Hand-Crafted Domain-Based Features for Efficient Security Resource Allocation on Attack Graphs”, *Journal of Information Security and Applications*, 2026. **Under Review***
- **Mohammad Ryiad Al-Eiadeh** and Mustafa Abdallah, “Genigraph: A Genetic-Based Novel Security Defense Resource Allocation Method for Interdependent Systems Modeled by Attack Graphs”, *Computers & Security*, 2024.
- **Mohammad Ryiad Al-Eiadeh** and Mustafa Abdallah, “PR-DRA: PageRank-based Defense Resource Allocation Methods for Securing Interdependent Systems Modeled by Attack Graphs,” *International Journal of Information Security*, 2025.
- **Mohammad Ryiad Al-Eiadeh** and Mustafa Abdallah, “CBDRA-IS: Centrality-Based Defense Resource Allocation for Securing Interdependent Systems,” *ACM Transactions on Privacy and Security (TOPS)*, 2025.
- **Mohammad Ryiad Al-Eiadeh** and Mustafa Abdallah, “Auto-Sec: Meta-Learning for Automated Efficient Security Resource Allocation on Attack Graphs,” *IEEE Transactions on Dependable and Secure Computing (TDSC)*, under revision, 2025.
- **Mohammad Ryiad Al-Eiadeh** and Mustafa Abdallah, “SAG-ERC: Securing Attack-Graph-based Systems through Node Embedding, Ranking, and Clustering,” *IEEE Conference on Secure and Trustworthy CyberInfrastructure for IoT and Microelectronics (SaTC)*.
- **Mohammad Ryiad Al-Eiadeh** and Mustafa Abdallah, “AARA-PR: Asset-Aware PageRank-Based Security Resource Allocation Method for Attack Graphs,” *IEEE Interdisciplinary Conference on Electrics and Computer (INTCEC)*, Chicago, IL, USA, 2024.

Future Research Directions

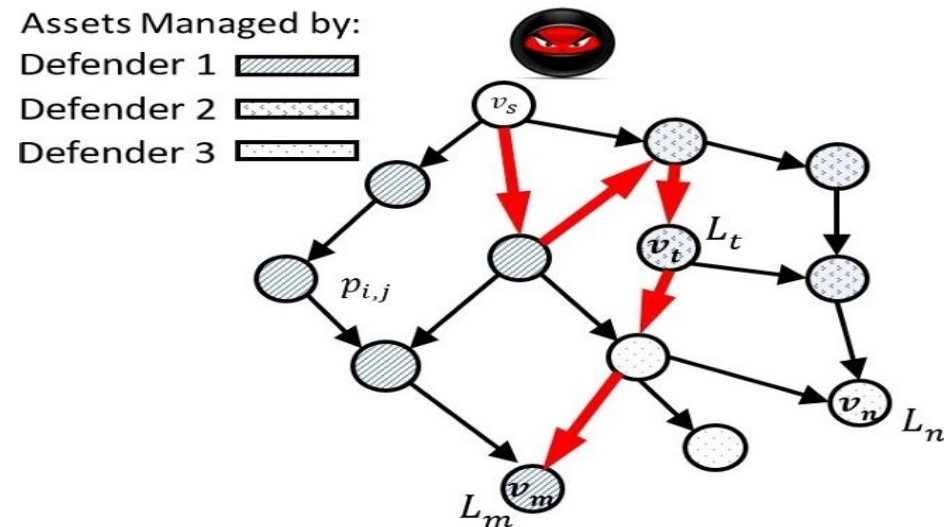
Evasive Adversarial Attack (stealthy & hard to detect)

- Investigate **evasive** adversarial attacks that use **gradient-like optimization** to **manipulate** node **embeddings** and **mislead** defender(s)
- Develop adversarial perturbation frameworks **targeting** embedding-based security allocation methods, such as RWs used in **Auto-Sec** and **SAG-ERC**



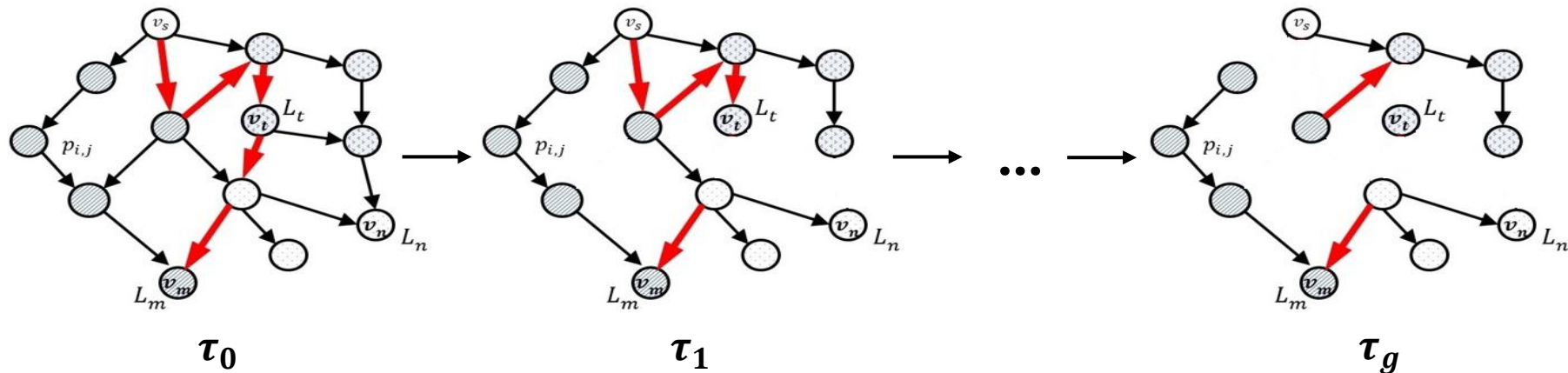
Multiple Defenders Based Clusters

- Extend SAG-ERC to **multi-defender** attack-graph settings and **measure** collective defense effectiveness using Cost-Relative Reduction (CRR) as a social cost reduction metric; **Clusters** are defined by **RWs**, ..., **GNN**
- Defenders work **under different settings** and **independently** at same time



Systems Evolve with Time (Dynamic Attack Graph)

- We can **incrementally** updates node embeddings using **Log-PPR** and **SVD** to efficiently **adapt** to evolving attack graphs
- The framework **reserves** resources for runtime **redistribution** and **reallocation** as vulnerabilities and network structures change
- **Questions:** 1) would it be efficient?! 2) How even to analyze dynamic attack graph?!



THANK YOU SO MUCH

Q & A

Appendix

Left empty by intent